

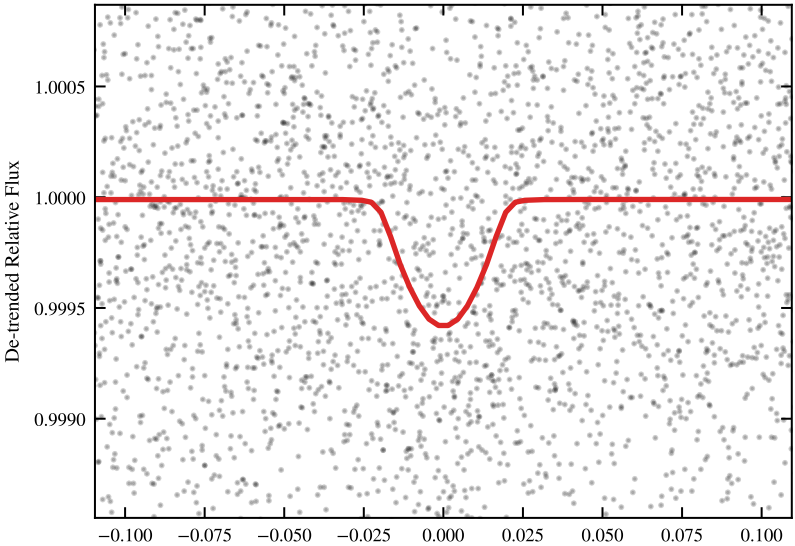
TESS PLANET TRANSIT CHARACTERIZATION REPORT

Target: TIC 101230735 | RA: 299.29100° | Dec: -48.93448° | Sectors: 104, 105

Derived Planetary Parameters

Parameter	Derived Value (MCMC)
Period (d)	2.0684 ± 0.00022
Rp/R★	0.038103 ± 0.0747
Rp (R⊕)	4.7534 ± 9.31
b	0.97464 ± 0.0916
T14 (h)	1.0509 ± 0.157
a (AU)	0.029898 ± 0.00353
Teq (K)	1691.2 ± 105

Primary Phased Transit



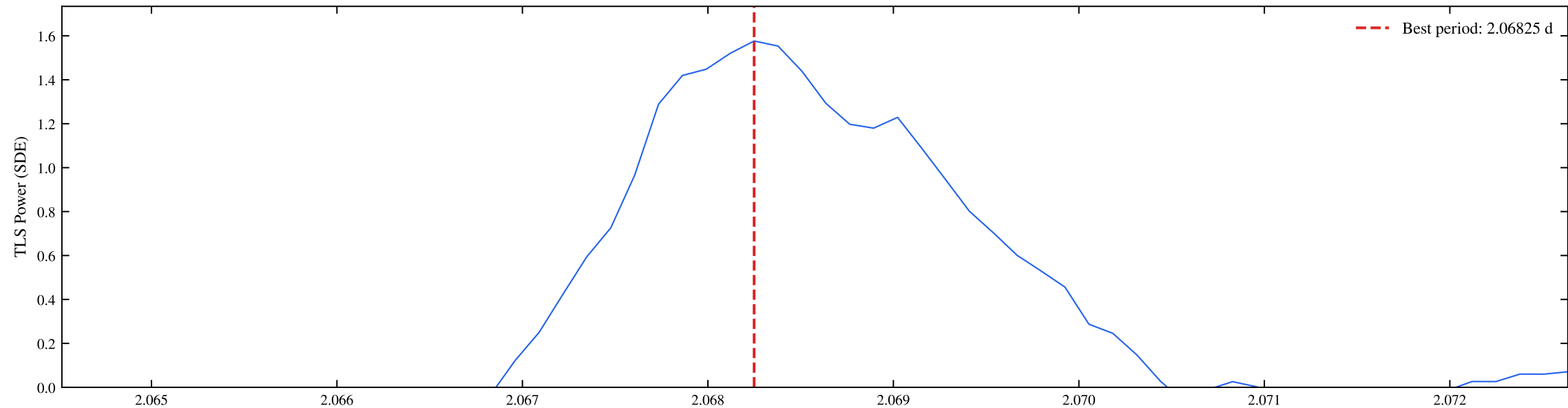
Literature Comparison & MCMC Diagnostics

Parameter / Metric	Literature Value	Derived / Diagnostic
P1 Period (d)	2.06825	2.068392 ± 0.00022
P1 Epoch (BTJD)	N/A	4179.6850 ± 0.0034
Max R-hat (R)	N/A	1.0020
Min ESS	N/A	2199
MCMC Divergences	N/A	5

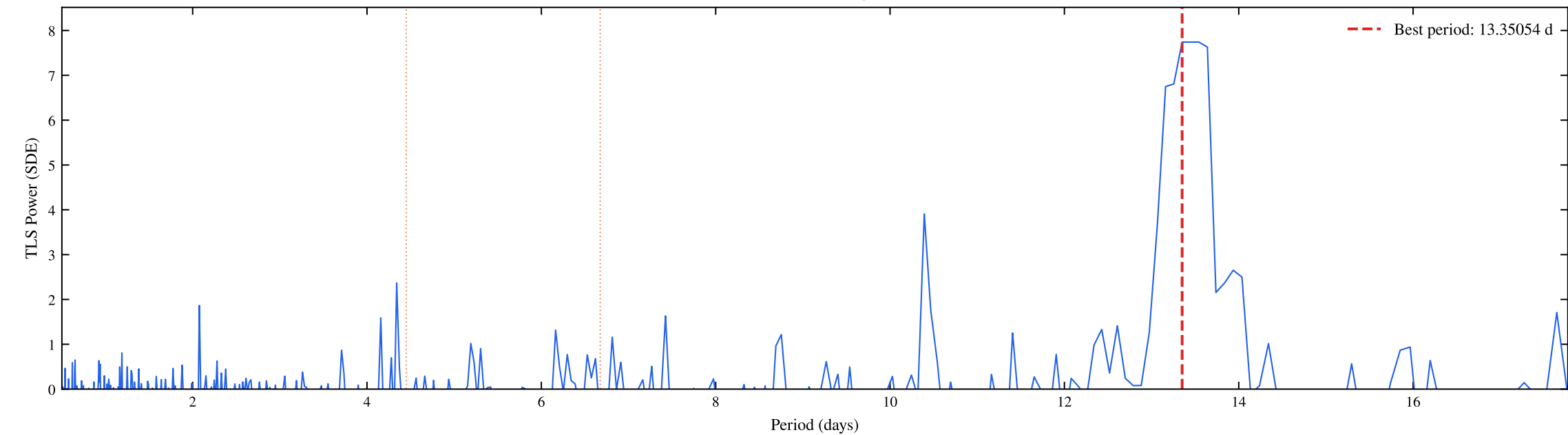
Host Star Properties

Stellar Parameter	Value	Source
R★ (R☉)	1.139 ± 0.064	VizieR TIC8
M★ (M☉)	1.01 ± 0.145	VizieR TIC8
Teff (K)	5688 ± 140	VizieR TIC8
log g (dex)	4.329 ± 0.082	VizieR TIC8
[Fe/H]	0.051	SIMBAD
ρ★ (g/cm³)	0.9637 ± 0.213	Derived

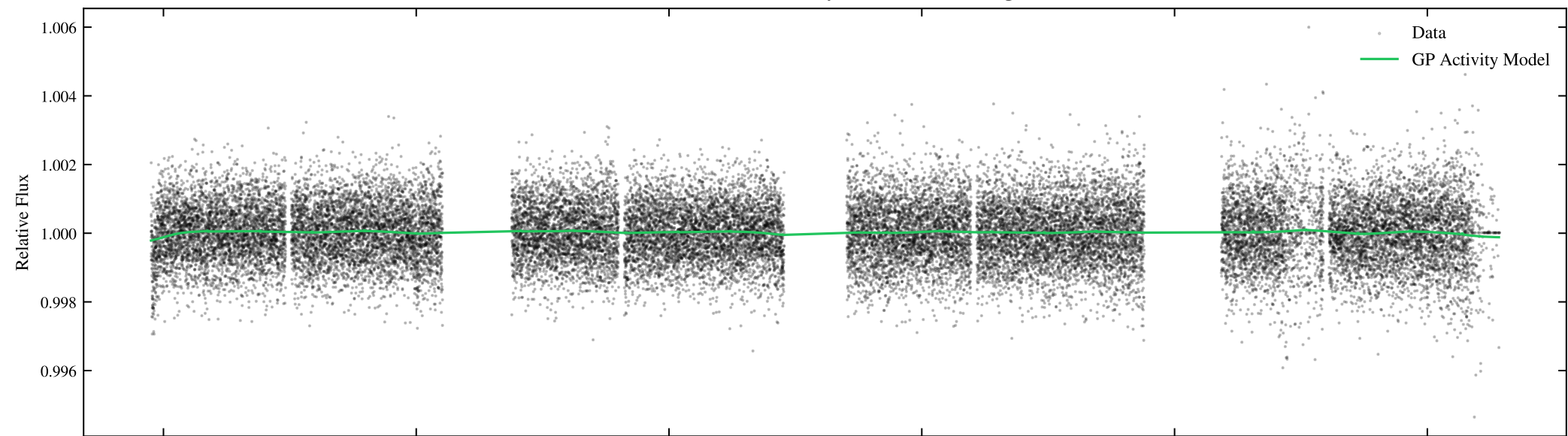
TIC 101230735 | Sectors: 104, 105 | TLS Periodogram (Planet 1 Search) (SDE = 1.58)



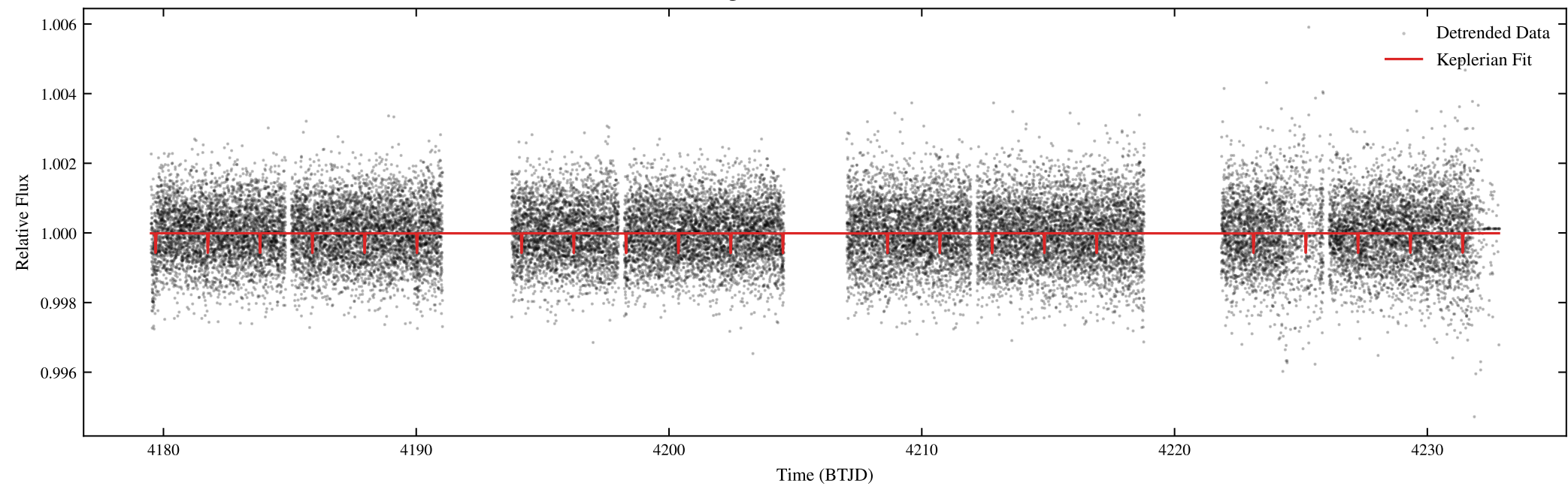
TIC 101230735 | Sectors: 104, 105 | TLS Periodogram (Planet 2 Search) (SDE = 7.74)



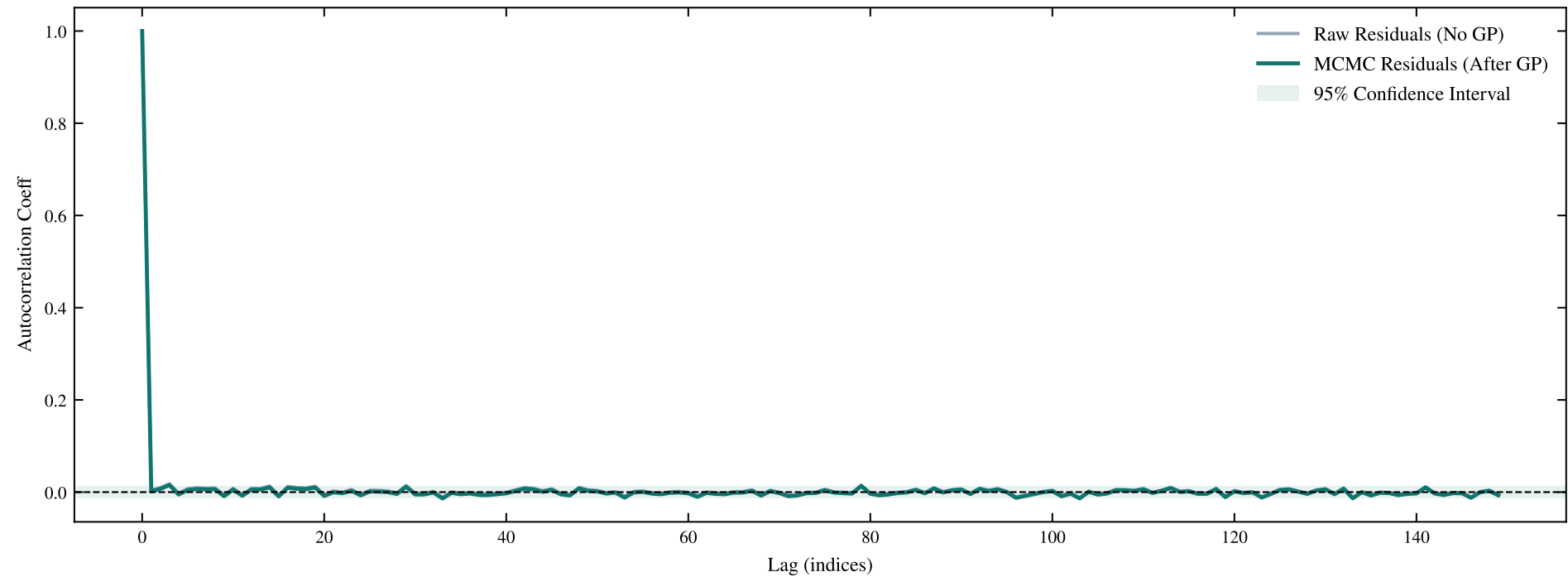
TIC 101230735 | Sectors: 104, 105 | Bayesian GP Detrending (Time Domain)



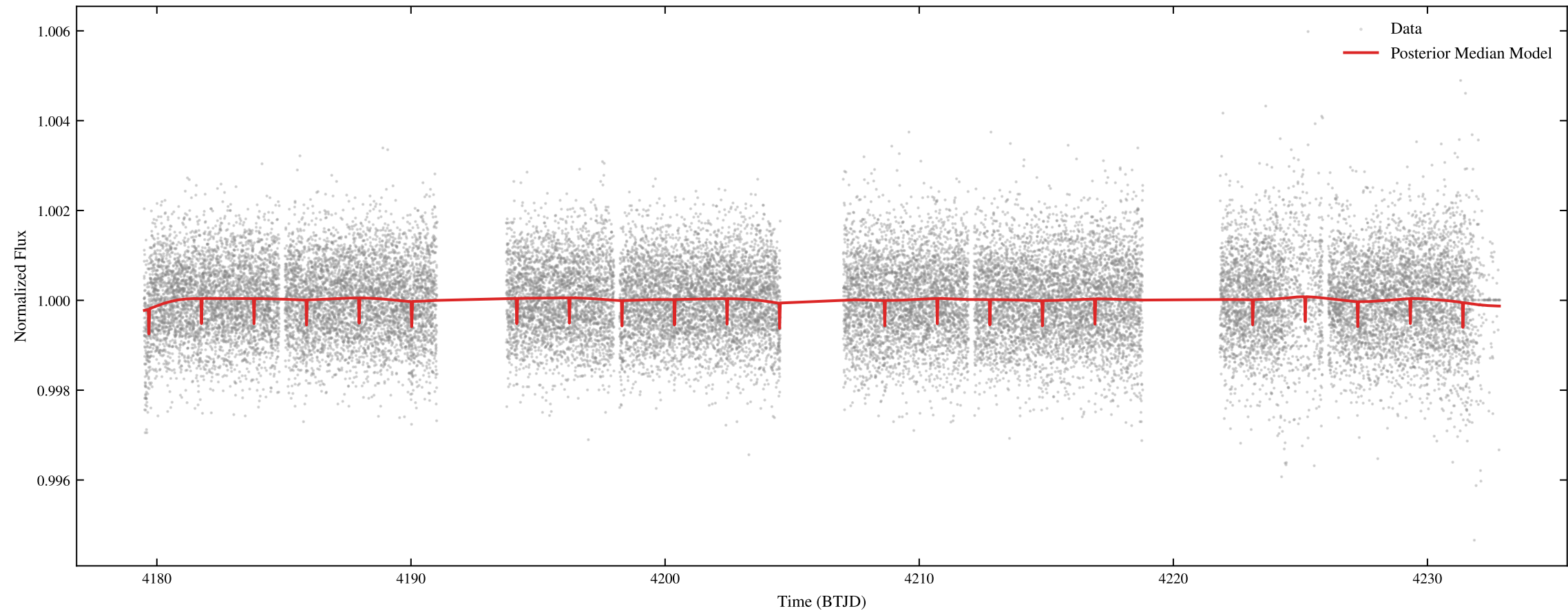
Detrended Light Curve & Best-Fit Transit Model

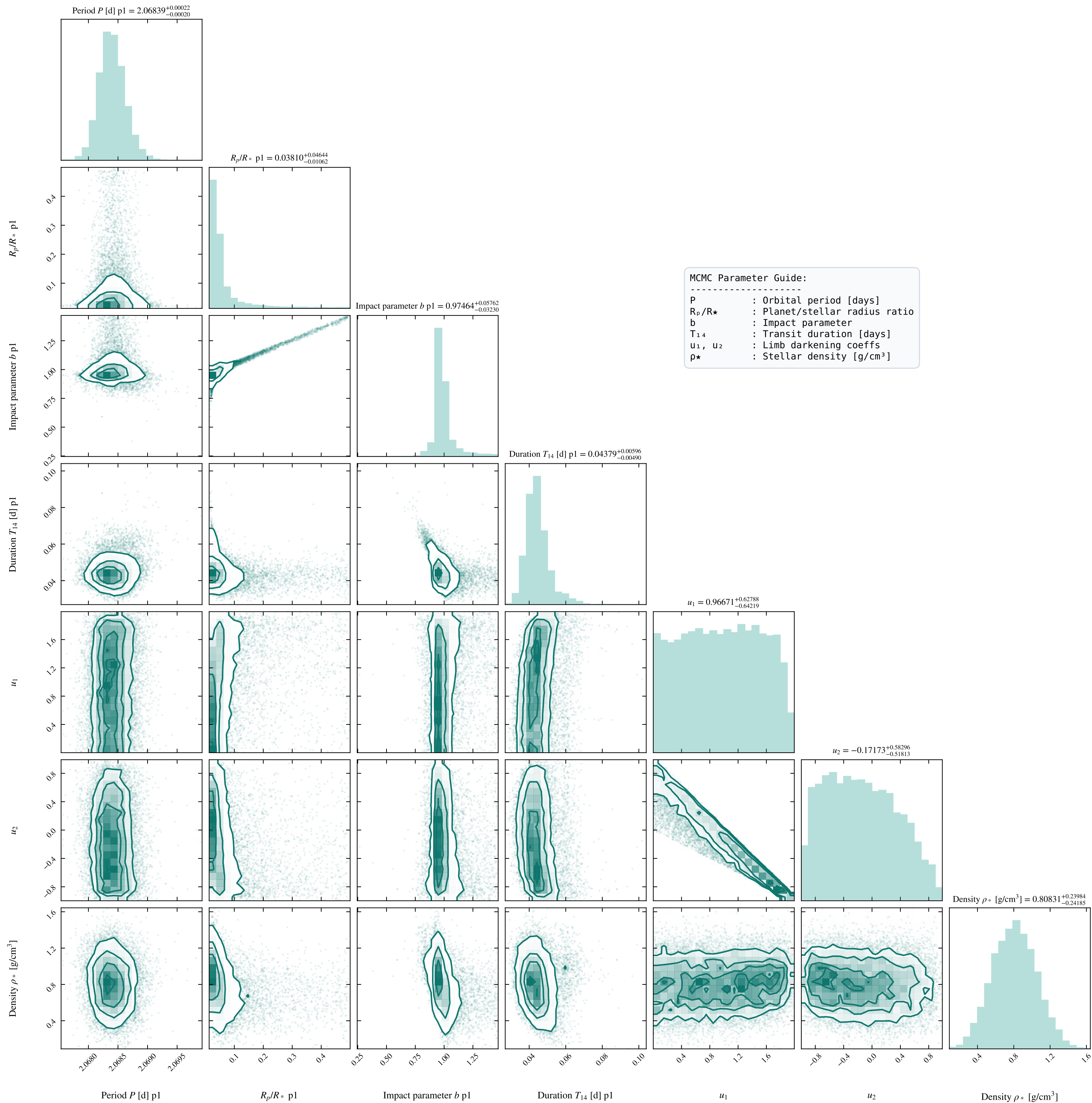


TIC 101230735 | Sectors: 104, 105 | GP Residuals Autocorrelation Function (ACF)

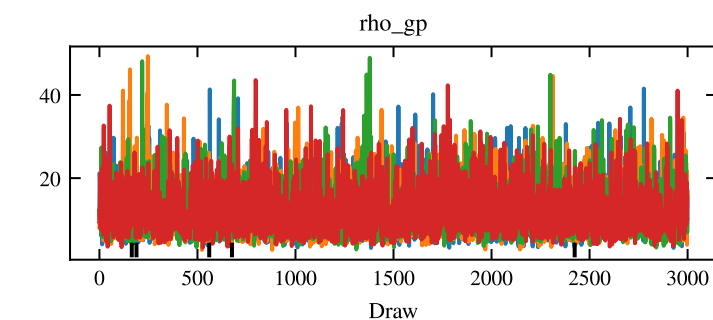
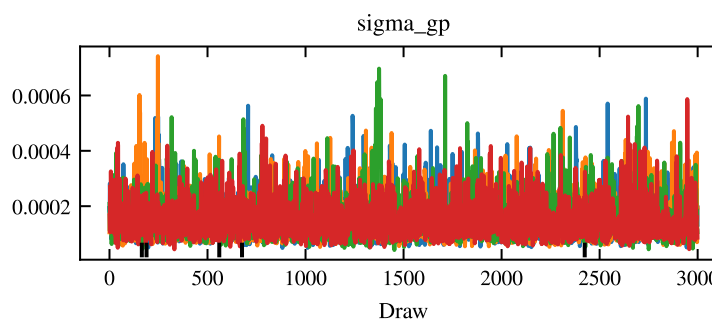
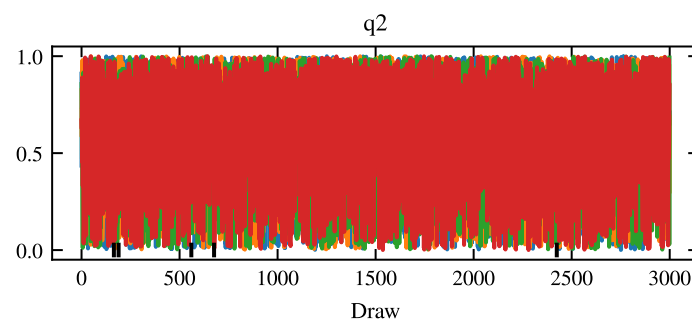
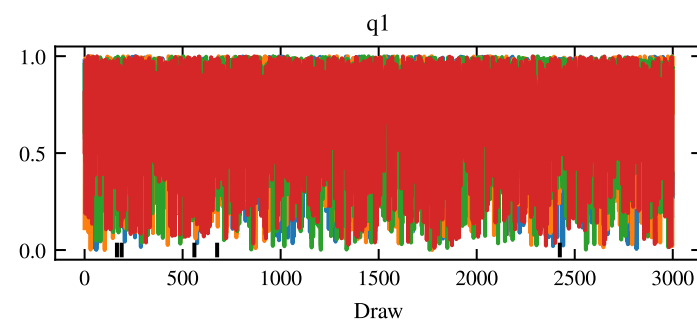
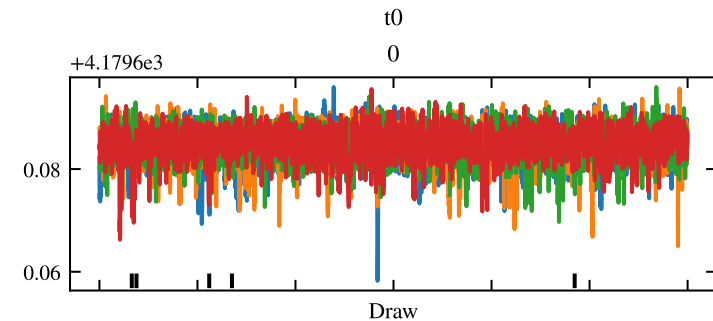
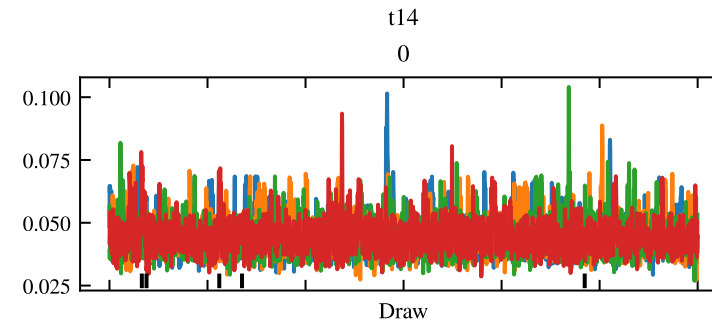
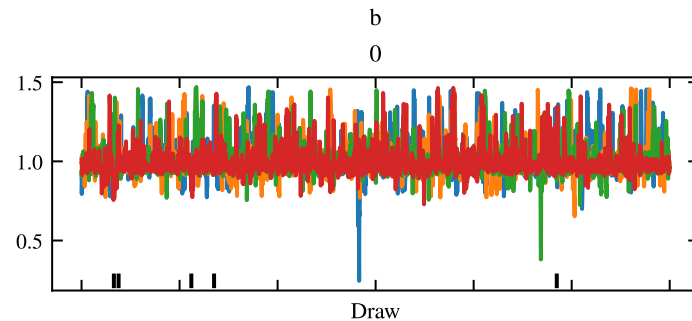
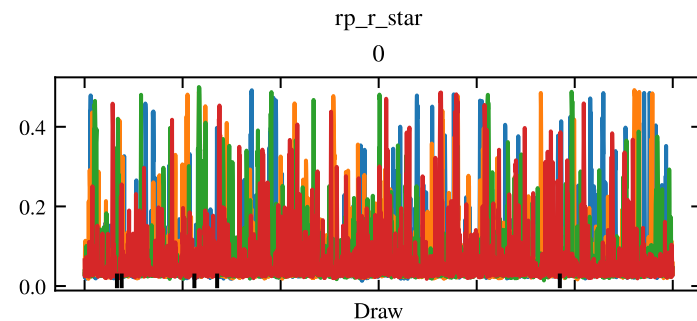


TIC 101230735 | Sectors: 104, 105 | MCMC Posterior Predictive Check

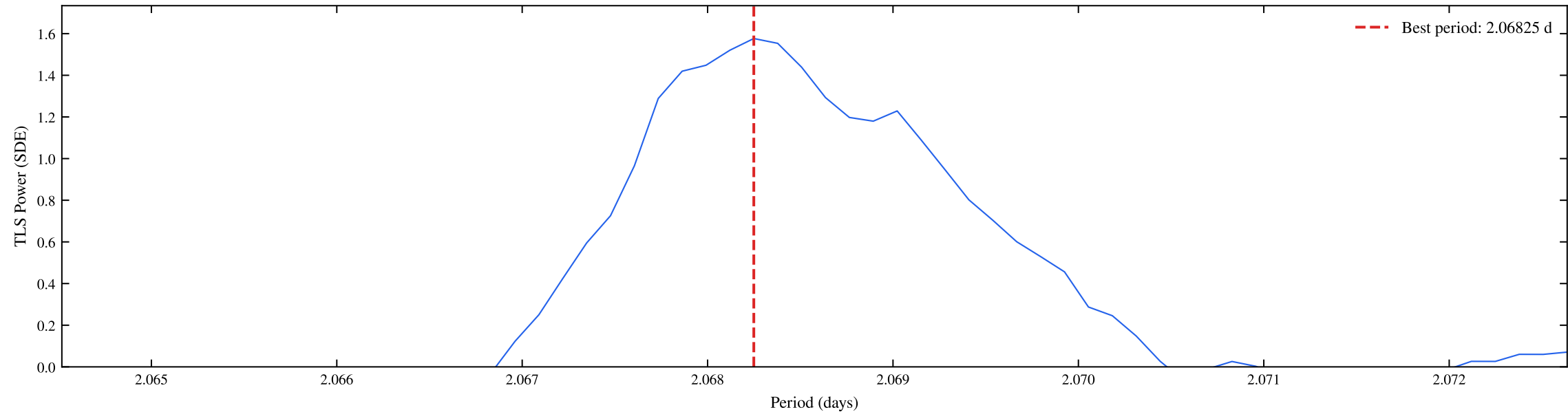




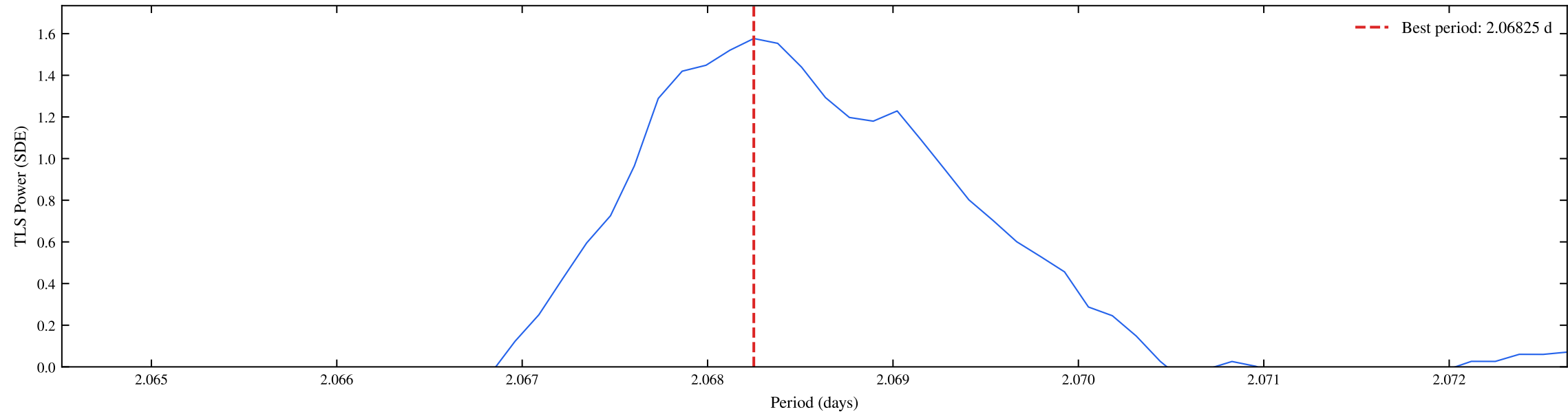
TIC 101230735 | Sectors: 104, 105 | MCMC Trace Plots



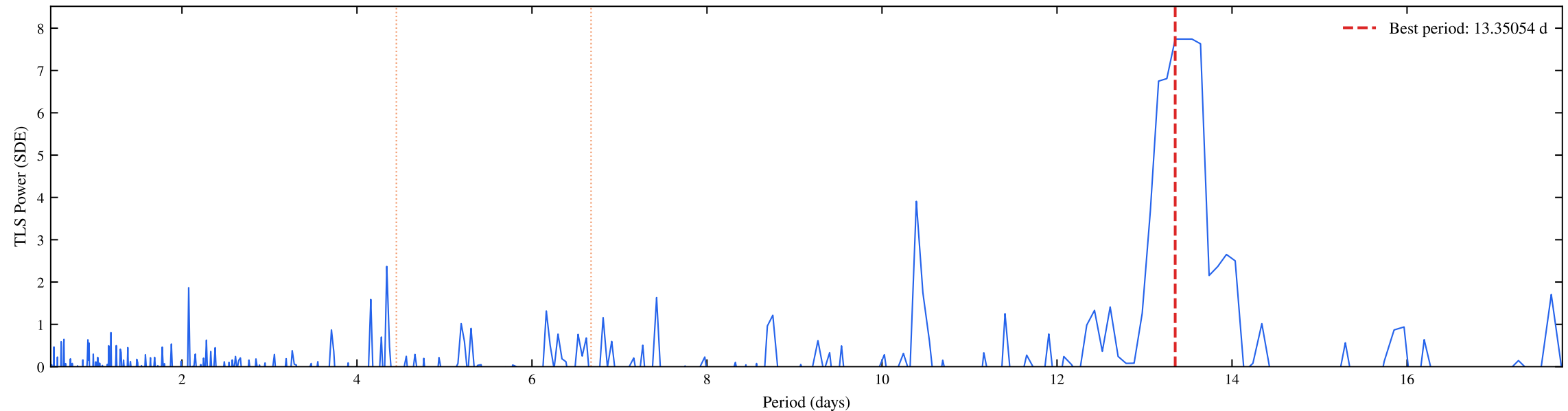
TIC 101230735 | Sectors: 104, 105 | TLS Coarse Periodogram (SDE = 1.58)



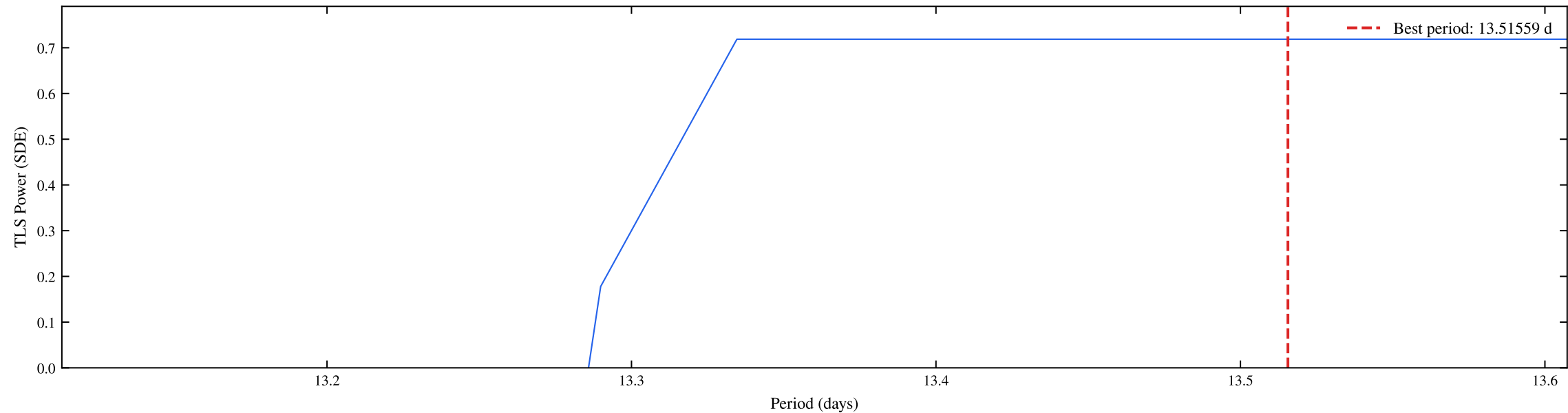
TIC 101230735 | Sectors: 104, 105 | TLS Fine Periodogram (SDE = 1.58)



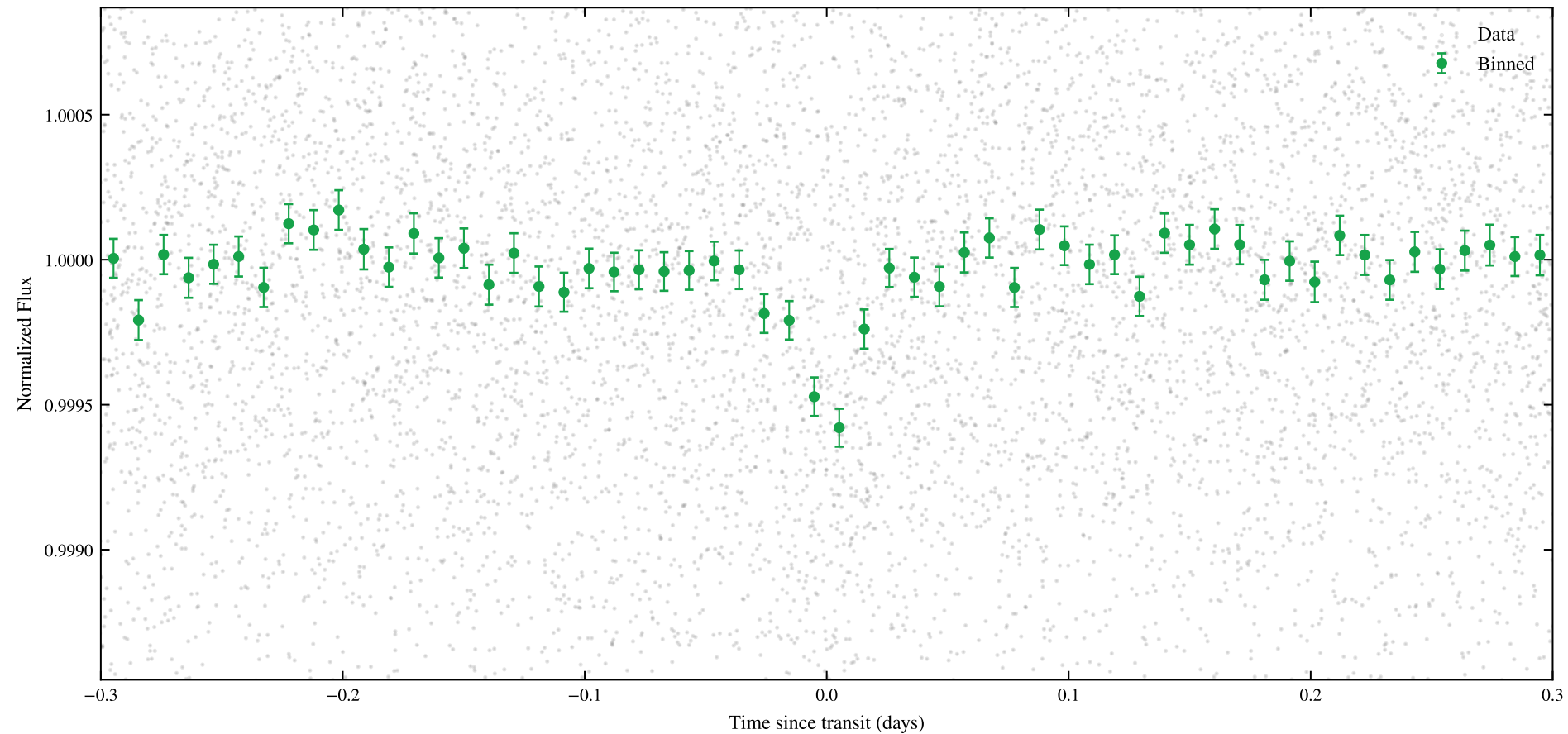
TIC 101230735 | Sectors: 104, 105 | TLS Coarse Periodogram (SDE = 7.74)



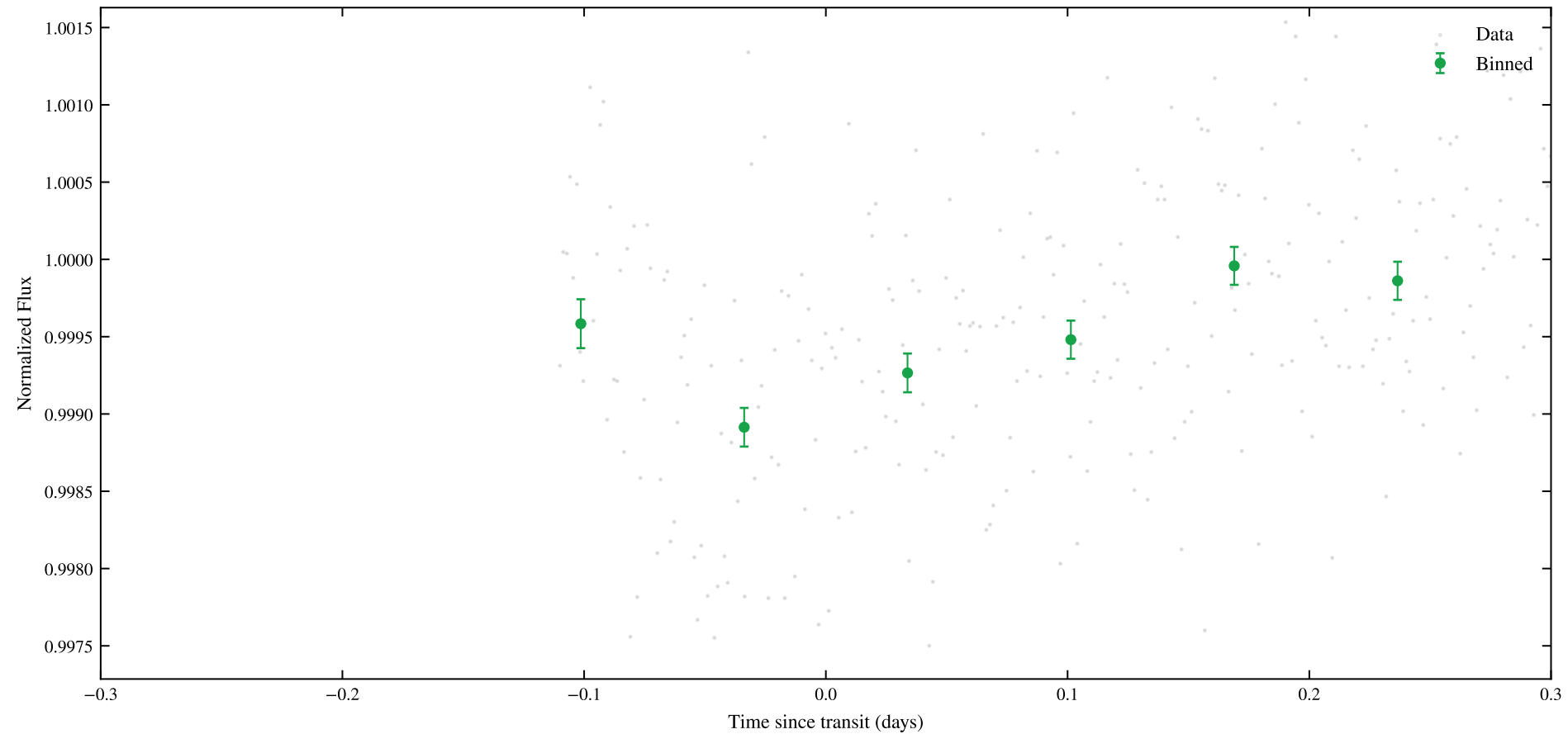
TIC 101230735 | Sectors: 104, 105 | TLS Fine Periodogram (SDE = 0.72)



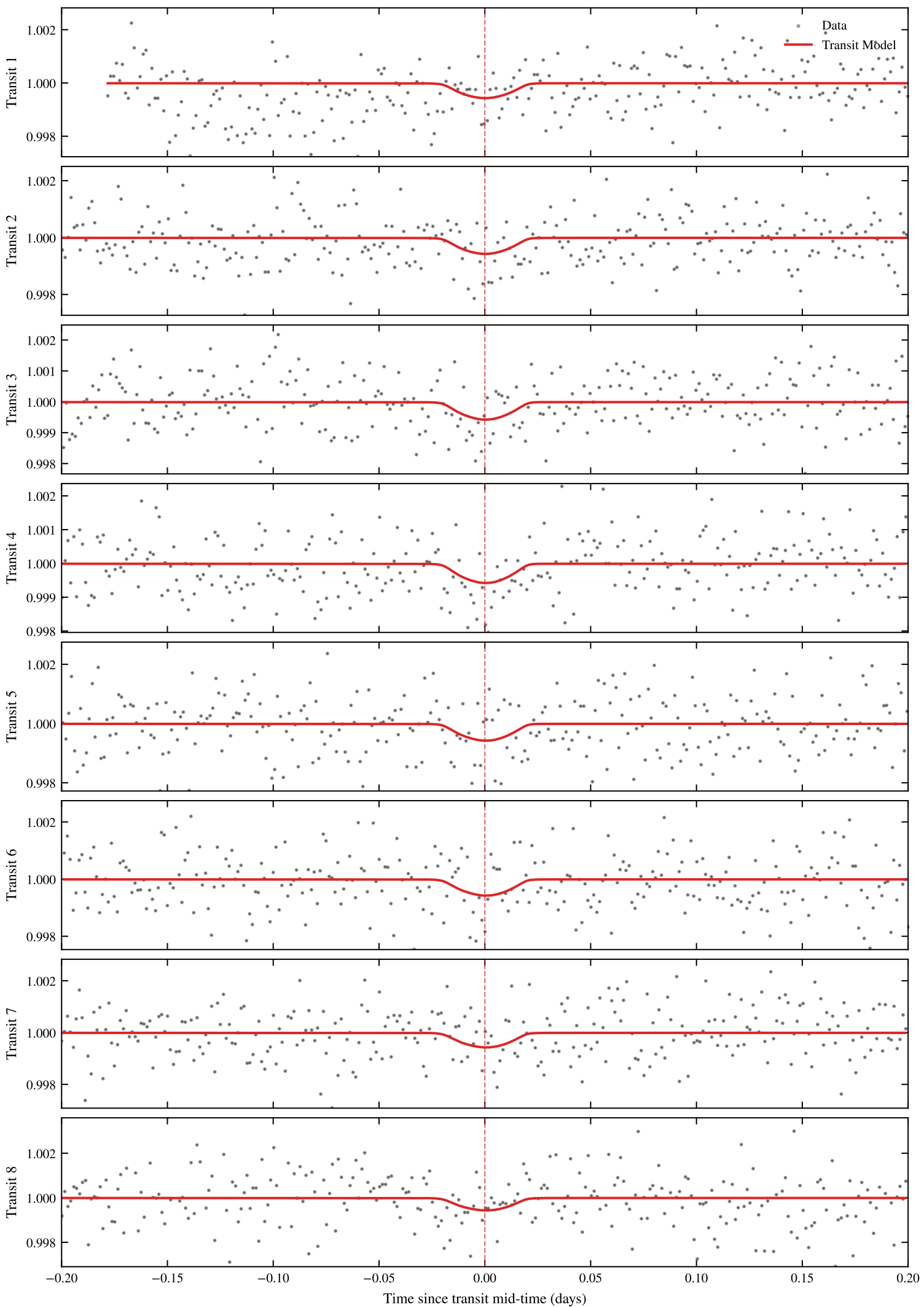
TIC 101230735 | Sectors: 104, 105 | Phase-Folded Transit (P = 2.06825 d)



TIC 101230735 | Sectors: 104, 105 | Phase-Folded Transit (P = 13.51559 d)



TIC 101230735 | Sectors: 104, 105 | Stacked Individual Transits



TIC 101230735 | Sectors: 104, 105 | Phased MCMC Fit - Planet 1

